

Boyan LI

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🔍 Google Scholar

Education

Hong Kong University of Science and Technology (Guangzhou)

Guangzhou, China

Ph.D. Student in Data Science and Analytics (DSA)

Sep. 2024 – Present

Supervisor: **Prof. Yuyu LUO**

Southern University of Science and Technology (SUSTech)

Shenzhen, China

B.Eng. in Computer Science and Technology

Sep. 2019 – Jul. 2023

GPA: 3.91/4.0 (Rank 2/183) | Honors: Outstanding Graduate, Highest Honors

Publications

* indicates equal contribution. **Boyan Li** is bolded.

First-Author Publications (Including Co-First)

- [1] **DataSpace: Benchmarking Data Agents for Verifiable Analytics over Heterogeneous Workspaces**
Boyan Li, Z. Liang, Y. Xie, X. Lin, T. Luo, X. Liu, Y. Zhu, Z. Peng, Y. Li, Z. Zhang, J. Zhang, N. Tang, G. Li, Y. Luo
arXiv preprint, 2026 [Paper]
Official evaluation benchmark for the KDD Cup 2026 Data Agents for Complex Data Analysis competition.
- [2] **DeepEye-SQL: A Software-Engineering-Inspired Text-to-SQL Framework**
Boyan Li, C. Chen, Z. Xue, Y. Mei, Y. Luo
ACM SIGMOD International Conference on Management of Data (SIGMOD), 2026 (CCF-A)
- [3] **DeepEye: A Steerable Self-driving Data Agent System**
Boyan Li, Y. Peng, Y. Xie, S. Lu, Y. Zhu, X. Mu, X. Liu, Y. Luo
ACM SIGMOD International Conference on Management of Data (SIGMOD), 2026 (Demo, CCF-A)
- [4] **DPC: Training-Free Text-to-SQL Candidate Selection via Dual-Paradigm Consistency**
Boyan Li, O. O. K. Hei, Y. Yu, Y. Luo
Annual Meeting of the Association for Computational Linguistics (ACL), 2026 (Main Conference, CCF-A)
- [5] **ROSE: An Intent-Centered Evaluation Metric for NL2SQL**
W. Pei*, S. Hou*, **Boyan Li***, H. Chen, Z. Shi, Y. Luo
Annual Meeting of the Association for Computational Linguistics (ACL), 2026 (Main Conference, CCF-A)
- [6] **Alpha-SQL: Zero-Shot Text-to-SQL using Monte Carlo Tree Search**
Boyan Li, J. Zhang, J. Fan, Y. Xu, C. Chen, N. Tang, Y. Luo
International Conference on Machine Learning (ICML), 2025 (CCF-A)
Deployed in **Huawei Cloud DataArts Insight** and **Wuhan Urban Operation Agent**.
- [7] **DeepVIS: Bridging Natural Language and Data Visualization Through Step-wise Reasoning**
Z. Shuai*, **Boyan Li***, S. Yan, Y. Luo, W. Yang
IEEE VIS Conference (VIS), 2025 (CCF-A)
- [8] **The Dawn of Natural Language to SQL: Are We Fully Ready?**
Boyan Li, Y. Luo, C. Chai, G. Li, N. Tang
Proceedings of the VLDB Endowment (VLDB), 2024 (CCF-A)
VLDB 2024 Top-2 Highly Cited Paper (Google Scholar Metrics).

Co-authored Publications

- [9] **MarkovSQL: Adaptive Text-to-SQL Workflow Orchestration Under Partial Observability**
Y. Zhu, **Boyan Li**, Y. Luo
ACM SIGMOD International Conference on Management of Data (SIGMOD), 2027 (Under Review)
- [10] **SQLConductor: Search-to-Policy Learning for Step-wise Text-to-SQL Orchestration**
Y. Zhu, Z. Peng, **Boyan Li**, Y. Luo
Proceedings of the VLDB Endowment (VLDB), 2027 (Under Review)
- [11] **DataMagic: Transforming Tabular Data into Data Insight Video**
Y. Xie, C. Ma, Z. Wang, L. Wang, J. Zhu, C. Zeng, Z. Shen, **Boyan Li**, Y. Luo
Proceedings of the VLDB Endowment (VLDB), 2026 (Demo Track, CCF-A)
- [12] **How to Make Tool-Using LLM Agents Efficient? A Systematic Survey**
Y. Hu, Z. Cheng, W. Pei, M. Li, **Boyan Li**, Y. Zhu, Y. Xie, H. Chen, S. Hou, Y. Luo, H. Cheng, J. X. Yu, H. Du
Conference on Language Modeling (COLM), 2026 (Under Review)
- [13] **A Survey of NL2SQL with Large Language Models: Where are we, and where are we going?**
X. Liu, S. Shen, **Boyan Li**, P. Ma, R. Jiang, Y. Zhang, J. Fan, G. Li, N. Tang, Y. Luo
IEEE Transactions on Knowledge and Data Engineering (TKDE), 2025 (CCF-A)
- [14] **nvBench 2.0: A Benchmark for Natural Language to Visualization under Ambiguity**
T. Luo, C. Huang, L. Shen, **Boyan Li**, S. Shen, W. Zeng, N. Tang, Y. Luo
Conference on Neural Information Processing Systems (NeurIPS), 2025 (CCF-A)

- [15] **NL2SQL-BUGs: A Benchmark for Detecting Semantic Errors in NL2SQL Translation**
X. Liu, S. Shen, **Boyan Li**, N. Tang, Y. Luo
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2025 (CCF-A)
- [16] **Augmenting Realistic Charts with Virtual Overlays**
Y. Shi, **Boyan Li**, Y. Luo, L. Chen, N. Tang
CHI Conference on Human Factors in Computing Systems (CHI), 2025 (CCF-A)
- [17] **EllieSQL: Cost-Efficient Text-to-SQL with Complexity-Aware Routing**
Y. Zhu, R. Jiang, **Boyan Li**, N. Tang, Y. Luo
Conference on Language Modeling (COLM), 2025
- [18] **A Survey of Data Agents: Emerging Paradigm or Overstated Hype?**
Y. Zhu, ..., **Boyan Li**, et al.
IEEE Transactions on Knowledge and Data Engineering (TKDE), 2025 (Under Review)
[Ranked Top-3 in Hugging Face Daily Papers.](#)
- [19] **Advances and Challenges in Foundation Agents**
B. Liu, ..., **Boyan Li**, et al.
arXiv preprint, 2025
[Ranked Top-1 in Hugging Face Daily Papers.](#)
- [20] **Are Large Language Models Good Statisticians?**
Y. Zhu, S. Du, **Boyan Li**, Y. Luo, N. Tang
Conference on Neural Information Processing Systems (NeurIPS), 2024 (CCF-A)
- [21] **A Plug-and-Play Natural Language Rewriter for Natural Language to SQL**
P. Ma, **Boyan Li**, R. Jiang, J. Fan, N. Tang, Y. Luo
arXiv preprint, 2024

Open Source & Systems

- **DeepEye: A Steerable Self-driving Data Agent System** [\[GitHub\]](#)
 - Led the development of a low-code orchestration platform for transparent data analysis.
 - **Awards:** “Best Open-source Project Award” at AI Agent 2025 Competition; Silver Medal at the 51st Geneva International Exhibition of Inventions.
- **Alpha-SQL (ICML 2025)** [\[GitHub\]](#)
 - Official implementation of the Monte Carlo Tree Search based Text-to-SQL agent.
 - **Impact:** Deployed in **Huawei Cloud DataArts Insight** and **Wuhan Urban Operation Agent**.
- **NL2SQL Handbook (TKDE 2025)** [\[GitHub\]](#)
 - A comprehensive and continuously updated roadmap for NL2SQL researchers.
 - **Impact:** Widely recognized as a seminal resource for tracking state-of-the-art advancements, with GitHub Stars 1.4k+.
- **NL2SQL360 Evaluation Framework (VLDB 2024)** [\[GitHub\]](#)
 - A comprehensive testbed for fine-grained evaluation of NL2SQL solutions.
 - **Recognition:** Associated with the VLDB 2024 Top-2 Highly Cited Paper.

Experience

Tsinghua University (Database Group)	<i>Beijing, China</i>
<i>Exchange Student (Mentor: Prof. Guoliang Li)</i>	<i>Jun. 2025 – Dec. 2025</i>
Huawei Central Research Institute	<i>Shenzhen, China</i>
<i>Research Intern</i>	<i>Jul. 2022 – Sep. 2022</i>

Honors & Awards

Merit Prize , INFH Research Excellence Award	<i>May. 2026</i>
Silver Medal , 51st Geneva International Exhibition of Inventions (DeepEye Data Agent System)	<i>Mar. 2026</i>
Silver Medal , 51st Geneva International Exhibition of Inventions (Low-Altitude Economy Ground-Air Collaborative Delivery Digital Twin Platform)	<i>Mar. 2026</i>
Research Excellence Awards , DSA Excellence Awards 2025	<i>Dec. 2025</i>
Runner-up, Best Project Award , DSAA 6000R, HKUST(GZ)	<i>Nov. 2025</i>
Best Open-source Project Award , AI Agent 2025 Competition	<i>Oct. 2025</i>
Merit Prize , DSA Excellent Research Award 2024	<i>Jan. 2025</i>
Best Project Award , DSAA 5021, HKUST(GZ)	<i>Dec. 2024</i>
Highest Honors in Computer Science & Engineering , SUSTech	<i>Jun. 2023</i>
Outstanding Graduate , SUSTech	<i>Jun. 2023</i>
Huawei Scholarship , Huawei Technologies	<i>May. 2022</i>
Excellent Student Scholarship , SUSTech	<i>2020, 2021, 2022</i>
National Encouragement Scholarship , Ministry of Education of China	<i>Apr. 2020</i>

Services

General Chair: Data Agents for Complex Data Analysis (KDD Cup 2026)	<i>2026</i>
Program Chair: Agentic Data Systems (VLDB 2026 Workshop)	<i>2026</i>

Conference Reviewer: Neural Information Processing Systems (**NeurIPS**); International Conference on Learning Representations (**ICLR**) 2026

Journal Reviewer: IEEE Transactions on Knowledge and Data Engineering (**TKDE**) 2026

Invited Talk: Huawei Squirrel Webinar, “DeepEye-SQL” Jan. 2026